
B Sets, Etc.

Many chapters of this book touch on the elements of discrete mathematics. This appendix reviews the notations, definitions, and elementary properties of sets, relations, functions, graphs, and trees. If you are already well versed in this material, you can probably just skim this chapter.

B.1 Sets

A *set* is a collection of distinguishable objects, called its *members* or *elements*. If an object x is a member of a set S , we write $x \in S$ (read “ x is a member of S ” or, more briefly, “ x belongs to S ”). If x is not a member of S , we write $x \notin S$. To describe a set explicitly, write its members as a list inside braces. For example, to define a set S to contain precisely the numbers 1, 2, and 3, write $S = \{1, 2, 3\}$. Since 2 belongs to the set S , we can write $2 \in S$, and since 4 is not a member, we can write $4 \notin S$. A set cannot contain the same object more than once,¹ and its elements are not ordered. Two sets A and B are *equal*, written $A = B$, if they contain the same elements. For example, $\{1, 2, 3, 1\} = \{1, 2, 3\} = \{3, 2, 1\}$.

We adopt special notations for frequently encountered sets:

- $\emptyset$ denotes the *empty set*, that is, the set containing no members.
- $\mathbb{Z}$ denotes the set of *integers*, that is, the set $\{\dots -2, -1, 0, 1, 2, \dots\}$.

- $\mathbb{R}$ denotes the set of *real numbers*.
- $\mathbb{N}$ denotes the set of *natural numbers*, that is, the set $\{0, 1, 2, \dots\}$.²

If all the elements of a set A are contained in a set B , that is, if $x \in A$ implies $x \in B$, then we write $A \subseteq B$ and say that A is a *subset* of B . A set A is a *proper subset* of set B , written $A \subset B$, if $A \subseteq B$ but $A \neq B$. (Some authors use the symbol “ $\subset$ ” to denote the ordinary subset relation, rather than the proper-subset relation.) Every set is a subset of itself: $A \subseteq A$ for any set A . For two sets A and B , we have $A = B$ if and only if $A \subseteq B$ and $B \subseteq A$. The subset relation is transitive (see page 1159): for any three sets A , B , and C , if $A \subseteq B$ and $B \subseteq C$, then $A \subseteq C$. The proper-subset relation is transitive as well. The empty set is a subset of all sets: for any set A , we have $\emptyset \subseteq A$.

Sets can be specified in terms of other sets. Given a set A , a set $B \subseteq A$ can be defined by stating a property that distinguishes the elements of B . For example, one way to define the set of even integers is $\{x : x \in \mathbb{Z} \text{ and } x/2 \text{ is an integer}\}$. The colon in this notation is read “such that.” (Some authors use a vertical bar in place of the colon.)

Given two sets A and B , *set operations* define new sets:

- The *intersection* of sets A and B is the set

$$A \cap B = \{x : x \in A \text{ and } x \in B\}.$$
- The *union* of sets A and B is the set

$$A \cup B = \{x : x \in A \text{ or } x \in B\}.$$
- The *difference* between two sets A and B is the set

$$A - B = \{x : x \in A \text{ and } x \notin B\}.$$

Set operations obey the following laws:

Empty set laws:

$$A \cap \emptyset = \emptyset,$$

$$A \cup \emptyset = A.$$

Idempotency laws:

$$A \cap A = A,$$

$$A \cup A = A.$$

Commutative laws:

$$A \cap B = B \cap A,$$

$$A \cup B = B \cup A.$$

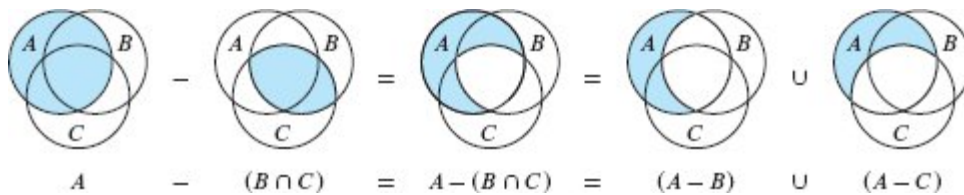


Figure B.1 A Venn diagram illustrating the first of DeMorgan's laws (B.2). Each of the sets A , B , and C is represented as a circle.

Associative laws:

$$A \cap (B \cap C) = (A \cap B) \cap C,$$

$$A \cup (B \cup C) = (A \cup B) \cup C.$$

Distributive laws:

$$\begin{aligned} A \cap (B \cup C) &= (A \cap B) \cup (A \cap C), \\ A \cup (B \cap C) &= (A \cup B) \cap (A \cup C). \end{aligned} \tag{B.1}$$

Absorption laws:

$$A \cap (A \cup B) = A,$$

$$A \cup (A \cap B) = A.$$

DeMorgan's laws:

$$\begin{aligned} A - (B \cap C) &= (A - B) \cup (A - C), \\ A - (B \cup C) &= (A - B) \cap (A - C). \end{aligned} \tag{B.2}$$

Figure B.1 illustrates the first of DeMorgan's laws, using a *Venn diagram*: a graphical picture in which sets are represented as regions of the plane.

Often, all the sets under consideration are subsets of some larger set U called the *universe*. For example, when considering various sets made up only of integers, the set $\mathbb{Z}$ of integers is an appropriate universe. Given a universe U , we define the *complement* of a set A as $\bar{A} = U - A = \{x : x \in U \text{ and } x \notin A\}$. For any set $A \subseteq U$, we have the following laws:

$$\begin{aligned}\bar{\bar{A}} &= A, \\ A \cap \bar{A} &= \emptyset, \\ A \cup \bar{A} &= U.\end{aligned}$$

An equivalent way to express DeMorgan's laws (B.2) uses set complements. For any two sets $B, C \subseteq U$, we have

$$\begin{aligned}\overline{B \cap C} &= \bar{B} \cup \bar{C}, \\ \overline{B \cup C} &= \bar{B} \cap \bar{C}.\end{aligned}$$

Two sets A and B are *disjoint* if they have no elements in common, that is, if $A \cap B = \emptyset$. A *collection* of sets $S_1, S_2, \dots$, either finite or infinite, is a set of sets, in which each member is a set S_i . A collection $\mathcal{S} = \{S_i\}$ of nonempty sets forms a *partition* of a set S if

- the sets are *pairwise disjoint*, that is, $S_i, S_j \in \mathcal{S}$ and $i \neq j$ imply $S_i \cap S_j = \emptyset$,
- their union is S , that is,

$$S = \bigcup_{S_i \in \mathcal{S}} S_i.$$

In other words, $\mathcal{S}$ forms a partition of S if each element of S appears in exactly one set $S_i \in \mathcal{S}$.

The number of elements in a set is the *cardinality* (or *size*) of the set, denoted $|S|$. Two sets have the same cardinality if their elements can be put into a one-to-one correspondence. The cardinality of the empty set

is $|\emptyset| = 0$. If the cardinality of a set is a natural number, the set is *finite*, and otherwise, it is *infinite*. An infinite set that can be put into a one-to-one correspondence with the natural numbers $\mathbb{N}$ is *countably infinite*, and otherwise, it is *uncountable*. For example, the integers $\mathbb{Z}$ are countable, but the reals $\mathbb{R}$ are uncountable.

For any two finite sets A and B , we have the identity

$$|A \cup B| = |A| + |B| - |A \cap B| , \quad (\text{B.3})$$

from which we can conclude that

$$|A \cup B| \leq |A| + |B|.$$

If A and B are disjoint, then $|A \cap B| = 0$ and thus $|A \cup B| = |A| + |B|$. If $A \subseteq B$, then $|A| \leq |B|$.

A finite set of n elements is sometimes called an *n -set*. A 1-set is called a *singleton*. A subset of k elements of a set is sometimes called a *k -subset*.

We denote the set of all subsets of a set S , including the empty set and S itself, by 2^S , called the *power set* of S . For example, $2^{\{a,b\}} = \{\emptyset, \{a\}, \{b\}, \{a, b\}\}$. The power set of a finite set S has cardinality $2^{|S|}$ (see Exercise B.1-5).

We sometimes care about setlike structures in which the elements are ordered. An *ordered pair* of two elements a and b is denoted (a, b) and is defined formally as the set $(a, b) = \{a, \{a, b\}\}$. Thus, the ordered pair (a, b) is *not* the same as the ordered pair (b, a) .

The *Cartesian product* of two sets A and B , denoted $A \times B$, is the set of all ordered pairs such that the first element of the pair is an element of A and the second is an element of B . More formally,

$$A \times B = \{(a, b) : a \in A \text{ and } b \in B\}.$$

For example, $\{a, b\} \times \{a, b, c\} = \{(a, a), (a, b), (a, c), (b, a), (b, b), (b, c)\}$. When A and B are finite sets, the cardinality of their Cartesian product is

$$|A \times B| = |A| \cdot |B| . \quad (\text{B.4})$$

The Cartesian product of n sets $A_1, A_2, \dots, A_n$ is the set of *n -tuples*

$$A_1 \times A_2 \times \dots \times A_n = \{(a_1, a_2, \dots, a_n) : a_i \in A_i \text{ for } i = 1, 2, \dots, n\},$$

whose cardinality is

$$|A_1 \times A_2 \times \dots \times A_n| = |A_1| \cdot |A_2| \cdot |A_n|$$

if all sets A_i are finite. We denote an n -fold Cartesian product over a single set A by the set

$$A^n = \underbrace{A \times A \times \dots \times A}_{n \text{ times}},$$

whose cardinality is $|A^n| = |A|^n$ if A is finite. We can also view an n -tuple as a finite sequence of length n (see page 1162).

Intervals are continuous sets of real numbers. We denote them with parentheses and/or brackets. Given real numbers a and b , the **closed interval** $[a, b]$ is the set $\{x \in \mathbb{R} : a \leq x \leq b\}$ of reals between a and b , including both a and b . (If $a > b$, this definition implies that $[a, b] = \emptyset$.) The **open interval** $(a, b) = \{x \in \mathbb{R} : a < x < b\}$ omits both of the endpoints from the set. There are two **half-open intervals** $[a, b) = \{x \in \mathbb{R} : a \leq x < b\}$ and $(a, b] = \{x \in \mathbb{R} : a < x \leq b\}$, each of which excludes one endpoint.

Intervals can also be defined on the integers by replacing $\mathbb{R}$ in the these definitions by $\mathbb{Z}$. Whether the interval is defined over the reals or integers can usually be inferred from context.

Exercises

B.1-1

Draw Venn diagrams that illustrate the first of the distributive laws (B.1).

B.1-2

Prove the generalization of DeMorgan's laws to any finite collection of sets:

$$\overline{A_1 \cap A_2 \cap \dots \cap A_n} = \overline{A_1} \cup \overline{A_2} \cup \dots \cup \overline{A_n},$$

$$\overline{A_1 \cup A_2 \cup \dots \cup A_n} = \overline{A_1} \cap \overline{A_2} \cap \dots \cap \overline{A_n}.$$

★ **B.1-3**

Prove the generalization of equation (B.3), which is called the *principle of inclusion and exclusion*:

$$\begin{aligned} |A_1 \cup A_2 \cup \dots \cup A_n| = & \\ & |A_1| + |A_2| + \dots + |A_n| \\ & - |A_1 \cap A_2| - |A_1 \cap A_3| - \dots \quad (\text{all pairs}) \\ & + |A_1 \cap A_2 \cap A_3| + \dots \quad (\text{all triples}) \\ & \vdots \\ & + (-1)^{n-1} |A_1 \cap A_2 \cap \dots \cap A_n|. \end{aligned}$$

B.1-4

Show that the set of odd natural numbers is countable.

B.1-5

Show that for any finite set S , the power set 2^S has $2^{|S|}$ elements (that is, there are $2^{|S|}$ distinct subsets of S).

B.1-6

Give an inductive definition for an n -tuple by extending the set-theoretic definition for an ordered pair.

B.2 Relations

A *binary relation* R on two sets A and B is a subset of the Cartesian product $A \times B$. If $(a, b) \in R$, we sometimes write $a R b$. When we say that R is a binary relation on a set A , we mean that R is a subset of $A \times A$. For example, the “less than” relation on the natural numbers is the set $\{(a, b) : a, b \in \mathbb{N} \text{ and } a < b\}$. An n -ary relation on sets $A_1, A_2, \dots, A_n$ is a subset of $A_1 \times A_2 \times \dots \times A_n$.

A binary relation $R \subseteq A \times A$ is *reflexive* if
 $a R a$